DESCRIPTIVE STATISTICS

Load the dataset ‘k_circles_sales’ as a csv file and read it using pandas

Get the basic information about data:

A) No. of rows and columns

B) Datatypes

C) Columns

D) Description

1 .Count values in item_fat_content column

2. Find unique values in item_identifier

3. Replace LF with Low Fat and reg with regular

4. Find the mean weight of all soft drinks

5. Find mean weight of items whose profit is more than 10.5

6. Find IQR of item_MRP

7. Find the 30th percentile of item_MRP.

8. Find variance, standard deviation, coefficient of variation of item_MRP

9. Find skewness and kurtosis of item_identifier

10. Prepare a histogram for item_type.

To replace the values

```
10.df.replace(to_replace="LF", value="Low Fat", inplace=True)
11.df_sales.replace(to_replace="reg", value="Regular", inplace=True)
```

To find mean

```
13.df[df['Item_Type'] == 'Soft Drinks' ]['Item_Weight'].mean()
14.df[df['profit']>10.5]['item_weight'].mean()
```

To find median and mode

```
16.df.select_dtypes(include="number").median()
17.df.median(numeric_only=True)
```

```
18.df.select_dtypes(include=object).mode()
```

```
26.## FILL MISSING VALUES WITH MEAN (numeric only)
```

```
df[col]=df[col].fillna(df[col].mean)
```

To find quantiles

```
23.df.quantile(0.25,numeric_only=true)  (Q1)
24.df.quantile(0.2,numeric_only=true)    (D2)
25.df.quantile(0.20,numeric_only=true)   (P20)
   IQR= df.quantile(0.75,numeric_only=true)-df.quantile(0.25,numeric_only=true)
```

```
28.##variance
df['item_Weight'].var()
df.var(numeric_only=True)
```

```
29.## standard deviation
df['item_Weight'].std()
df.std(numeric_only=True)
```

```
30.coefficient of variation
df['item_weight'].variation()
31.df['item_weight'].skew()
        df.skew(numeric_only=True)
32.df['item_weight'].kurt()
        df.kurt(numeric_only)
```

```
33.## for creating normal distribution
data= np.random.normal(50,10,50)
mean = np.mean(data)
median = np.median(data)
mode = stats.mode(data)[0]

34. ## for creating skewed data
data1= np.random.normal(50,10,50)**5
mean = np.mean(data1)
median = np.median(data1)
mode = stats.mode(data1)[0]
```